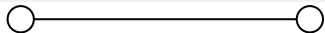


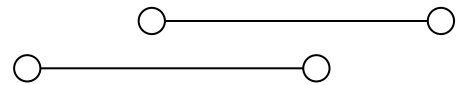


WatMatch

Creating Canada's Largest Capstone Ecosystem

Siddharth, Samson, Ayman, Darsh, Sparsh

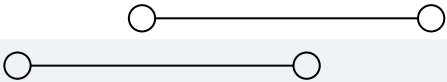




Creating Canada's Largest Capstone Ecosystem

- Global Futures Initiative (more info [here](#))
- A [Digital Home](#) for Capstone Partner Collaboration
- Our contacts:
 - Prof. Derek Rayside (Initiative Lead, Mentor)
 - Mr. Jamieson Cox (Centre for Work-Integrated Learning)
- We're first rolling out this initiative to UWaterloo students
- Long-term vision: accessible to all Canadian Universities





✦ 01 ✦

Problem Recap



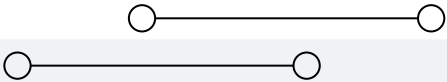
Manual Matching Work

Current cross-faculty matching depends on large PDFs built from student survey responses

4 main problems

1. Students cannot easily search/compare opportunities
2. Most students form teams with people they already know
3. Faculty/staff coordinate matching by hand
4. No clear system of record for consent, approvals, or decisions for interdisciplinary matching





✦ 02 ✦

Why this matters



Who does this impact?

A platform for capstone matching helps...

Students

Find teammates, projects, external opportunities beyond their own network!

Instructors

Review proposals, give feedback, manage teams, track decisions in one centralized roster!

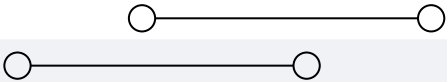
External Partners

Share opportunities and ideas with students!

Universities

To support interdisciplinary capstones at a large scale





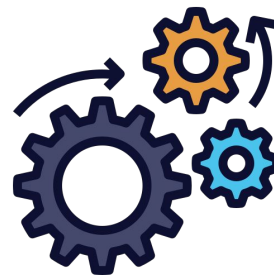
✦ 03 ✦

Final Status S26



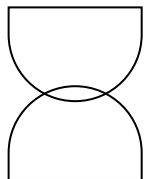


Final - Features Implemented



Final - Multi-Course WatMatch + Interdisciplinary Capstones

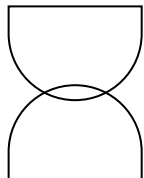
- ☐ Feature #10: Reason-aware audit log for capstone lifecycle
- ☐ Feature #11: Multi-Course WatMatch + Academic / Enrollment Operator workspaces
- ☐ Feature #12: Interdisciplinary routing ([Future Cities](#), [Global Engagement](#), GENE/SCI/ARTS)
- ☐ Feature #13: Marketplace Phase Implementation
 - 3 Phases: Exploration → commitment → finalization
 - Globally shared for interdisciplinary courses, instructor-scoped for regular courses
- ☐ Feature #14: Term turnovers through course pipelines (ECE 498A → ECE 498B)
- ☐ Feature #15: Completed WatMatch-based capstones → Past Capstones
- ☐ Stakeholder Feedback: Role-separated student, instructor, mentor, etc... workspaces
- ☐ UI improvements for: <https://watmatch.vercel.app/>

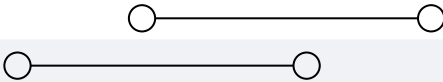




Final - Technical Design Decisions

Decision	Alternative Rejected	Reason
One default 3-phase timeline Admin exceptions for any course (example: SE 490) to override the phase dates	Let every project advance independently as they choose	Students in an interdisciplinary capstone should move together “globally”, but some courses (SE 490) won’t adhere to this global schedule
Upon capstone completion: publish completed snapshot & close the active project	Leave completed projects editable in the active marketplace	Past Capstones need to preserve the completed result without leaving project open
Enforce role and course access in the API and database (RBAC)	Only hide unauthorized buttons in the UI	Security risk - users should not bypass the interface and perform restricted actions through a direct API request
Record routing decision + Manually complete official enrollment in Quest	Silently replace course values or claim the student is already enrolled	Accountability: WatMatch records the coordinating course, each student’s intended route, who approved it, and why; Quest remains the official enrollment system





✦ 04 ✦

WatMatch Design





Final - Marketplace Design



Phase 1: Exploration

- Students save, express interest, accept invites, work on multiple projects at the same time
- Idea: students want to explore skillset/passion fit (no official membership yet)

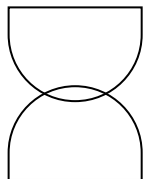
Phase 2: Commitment

- Existing student matches (from exploration) mutually confirm commitment
- Team leader proposes the complete roster to instructor during capstone approval

Phase 3: Finalization

- Any new marketplace activity stops during this phase
- Remaining unofficial memberships & blockers are resolved by instructors (or admins)
- Ready teams finalize (finalized rosters become unmodifiable by students)

Existing workflows from [Interdisciplinary Capstones](#) use similar ideas in how things are done!



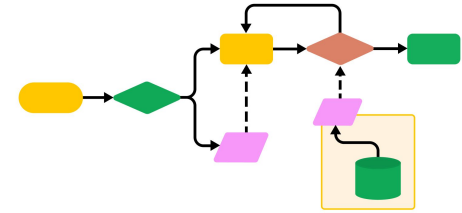


Final - Interdisciplinary Course Routing



Once a leader proposes a capstone to their instructor

WatMatch asks “**Can every student be in one active course?**”

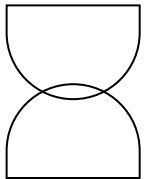


If **YES**: Same Course route

- Apply official membership + Return to the instructor for review
- Example: Student 1 (SYDE 461) + Student 2 (SYDE 461)

If **NO**: Mixed / Interdisciplinary Course route

- Academic Advisor / Enrollment Operator routes them appropriately (per student)
- Return to instructor for review
- Update students' Quest manually outside WatMatch
- Example: Student 1 (ARTS 450) + Student 2 (GENE 403) keep separate enrollment routes





Final - Interdisciplinary Offerings



Standalone vs Held-With offerings

Standalone Course: [SYDE 461 \(Fall 2026\)](#)

- One course offering + No “Held With” entries at all

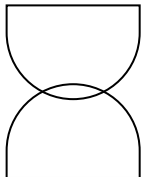
Shared Course: [ARTS 450 \(Fall 2026\)](#)

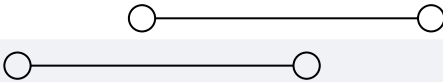
- Held with GENE 403, MTHEL 398, SCI 300, etc...
- Only applies to F26, not permanently set-in-stone

Subject	Catalog#	Units	
ARTS	450	0.5	
Notes: Consult Global Engagement Program website for more information			
	Class Comp Sec Camp Loc Assoc. Class Rel 1 Rel 2		
	4180	SEM 001	UW U 1
	Held With: ENVS 403A PRJ 001		
	Held With: ENVS 404 PRJ 001		
	Held With: GENE 403 PRJ 001		
	Held With: HEALTH 490 SEM 001		
	Held With: MTHEL 398 LEC 001		
	Held With: SCI 300 LEC 001		
	Held With: SFM 403 LEC 001		
	Held With: SFM 413 LEC 001		

We let students share “**one capstone experience**” while keeping their own transcript course

Note: This group may be different in another term, so WatMatch stores **held-with** relationships as part of a term configuration (instead of permanently attaching to a course)





05

Stakeholder Engagement





Final - Meeting with Stakeholders

[08/06] Meeting with **Prof. Derek Rayside** and **Mr. Jamieson Cox**

What I presented

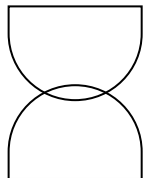
- Exploration → Commitment → Finalization → Term Completion → Past Capstones
- Multi-course routing: same-course teams vs interdisciplinary teams + enrollment operator
- Role-specific workspaces + new audit logging feature + team readiness guardrails

Feedback

- Fall 2026 Pilot should focus on SE 490 (still in discussion, subject to change)
- Plan to have multi-role accounts (example: Derek can be Instructor + University Mentor)
- Add an optional University Relationship Manager (only for [FCI capstones](#), this is rare)

Requests made to Prof. Rayside

- Pushed again for VM allocation from IST / ENG Computing (blocker for UW ADFS auth)
- Requested them to share <https://watmatch.vercel.app/> with their network of students/staff





Final - Meeting with Academic Advisor

[08/06] Meeting with Ms. Angie Hildebrand

What I presented

- Routing queues for mixed-course / interdisciplinary / no course
- Discussed Academic Advisor decisions + **role-boundaries**

Feedback

- Keep the user-facing name Academic Advisor
- UI is great! Shows context + looks appropriate
- Showing student email is enough info to work with
- **Adoption risk:** recording the decision in WatMatch after updating Quest feels like an extra step

ACADEMIC ADVISING

Advising workbench

• Fall 2026 · Exploration

Commitment routing

Confirm the review course and an enrollment route for each roster member.

Mixed-Course Sensor Platform

Recommended coordinator: MTE 481 - Mechatronics Engineering Design Project

1 pending

2 roster members

Team #280

student.se.mixed.leader@uwaterloo.ca Leader

Student context and secondary actions

Enrollment course

SE 490 - Design Project 1

student.mte.commit@uwaterloo.ca Pending commitment

Student context and secondary actions

Enrollment course

MTE 481 - Mechatronics Engineering Design Project

Coordinating review course

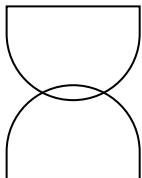
MTE 481 - Mechatronics Engineering Design Project

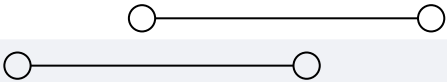
Decision notes

Required for route changes, returns, or rejection

Approval routes the complete roster together, assigns the selected enrollment courses, and returns the project to instructor review.

Approve routing





06

Final
Demo



Final - Student Leader Workspace

WotMatch

STUDENT WORKSPACE

Project details

A same-course SE marketplace project where the final candidate is ready for leader confirmation.

Finalize team

Reopen for edits

Invite teammate

More actions

TEAM MEMBERS

S

student.se.direct.leader@uwaterloo.ca

SE 490 - Design Project 1
Software Engineering

View profile

Finalization readiness

4/6 ready

Required support missing

An accepted mentor or confirmed external partner is still required.

Commitments need resolution

Resolve mutually confirmed explorations before finalization.

Instructor approval complete

Instructor review has approved this project.

No pending staff routing

No commitment routing is waiting on staff.

Team identity valid

The official roster includes a team leader.

Official enrollment recorded

Every official member has an active, staffed enrollment course.

Project support

Support required

An accepted mentor or confirmed external partner is still required.

Manage mentor support

EXPLORING STUDENTS

student.se.direct.candidate@uwaterloo.ca

Exploring

Course: SE 490 - Design Project 1
Home department: Software Engineering
Exploring since Aug 6, 2:36 PM - Updated Aug 6, 2:36 PM
"Both the candidate and leader have committed; leader still needs to confirm the final roster."

Stop exploring

Commitment confirmed

Commitment roster proposal

Propose all mutually confirmed candidates for course routing. Same-course proposals go straight to instructor review; mixed or no-course proposals go to staff routing.

Send roster proposal

PENDING INVITATIONS

Student profile

Profile details for student.se.direct.leader@uwaterloo.ca.

S

student.se.direct.leader@uwaterloo.ca

SE 490 - Design Project 1 - Software Engineering

Same-course leader

Afternoons

ABOUT

Building a practical scheduling tool for student teams.

SKILLS

Next.js

SQL

Leadership

PREFERRED ROLES

Team lead

PROJECT INTERESTS

Scheduling

Student life

INTERESTED DISCIPLINES

Software Engineering

LINKS

GitHub



Final - Enrollment Operator Workspace



ENROLLMENT OPERATIONS

Enrollment workbench

Fall 2026 - Exploration

Exploration
Sep 8

Commitment
Sep 22

Finalization
Oct 6

Commitments

1

Enrollment requests

1

Course routing

1

Courses needing
attention
48

Cycle settings
Exploration

Search projects or students

All courses

All departments

Refresh

6 waiting · oldest Aug 6, 2:36 PM

Mixed-Course Sensor Platform

Commitment Routing

student.mte.commit@uwaterloo.ca waiting since Aug 6, 2:36 PM

Future Cities Stormwater Commons

Course Reassignment

student.futurecities@uwaterloo.ca waiting since Aug 6, 2:36 PM

Submission enrollment

Project Submission Enrollment

student.nocourse@uwaterloo.ca waiting since Aug 6, 2:36 PM

Commitment routing

Confirm the review course and an enrollment route for each roster member.

Mixed-Course Sensor Platform

1 pending

2 roster members

Recommended coordinator: MTE 481 - Mechatronics Engineering Design Project

Team #280

student.se.mixed.leader@uwaterloo.ca

Leader

Enrollment course

Student context and secondary actions

SE 490 - Design Project 1

student.mte.commit@uwaterloo.ca

Pending commitment

Enrollment course

Student context and secondary actions

MTE 481 - Mechatronics Engineering Design Project

Coordinating review course

Decision notes

MTE 481 - Mechatronics Engineering Design Project

Required for route changes, returns, or rejection

Approval routes the complete roster together, assigns the selected enrollment courses, and returns the project to instructor review.

Approve routing



Final - Instructor Workspace



INSTRUCTOR • SE 490 - DESIGN PROJECT 1

Course workspace

[Course home](#) [Reviews](#) [Teams](#) [Roster](#)

More

Capstone teams

6 of 6 teams shown. Open a row for roster, readiness, support, and staff actions.

Recruiting SE 490

Unsupported Finalization Blocker

Team 281 • 1 official member • 0 pending invites

NEXT STAFF TASK
Resolve 1 finalization readiness item.
Readiness 5/6

COORDINATING COURSE

SE 490 - Design Project 1

Software Engineering Engineering Departmental

MEMBERS

 student.se.unsupported@uwaterloo.ca
SE 490 - Design Project 1
Software Engineering

PENDING INVITES

No pending invites.

Finalization readiness

5/6 ready

- ☒ Required support satisfied
Attach an accepted mentor or confirmed external partner first.
- ☒ Capstone approved for recruiting
Instructor review has approved this project for recruiting.
- ☒ No pending staff routing
No final commitment routing is waiting on staff.
- ☒ No unresolved confirmed explorations
No mutually confirmed exploration is still awaiting commitment resolution.
- ☒ Team identity valid
The team has confirmed members and a leader.
- ☒ Official enrollment courses valid
Every official member has an active staffed enrollment course.

PROJECT SUPPORT

- ☒ Support required
An accepted mentor or confirmed external partner is still required.

Select mentor

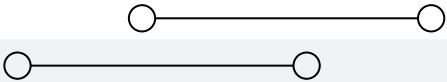
Optional note

Request Mentor

Select mentor

Dr. Alex Lee - Software Engineering - Product scoping, Accessibility

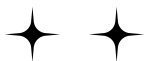
Staff actions



✦ 07 ✦

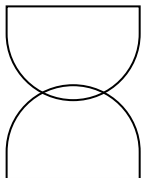
Plan for Fall 2026





SE 490 - F26 Development Plan

- **Feature #1:** Multi-role accounts + role-switching
 - One person can hold multiple roles without scoped accounts
 - Example: Derek switches between Instructor & University Mentor
- **Feature #2:** Course-scoped phase control (if admin approves standalone course)
 - An instructor can advance their assigned standalone course through 3 Phases
- **Feature #3:** Mentors / Instructors / TAs record milestone feedback on student capstones
- **Feature #4:** Action-items inbox + email notifications (for invites, commitment confirmations, routing decisions, instructor reviews, mentor requests, milestone deadlines, etc...)
- **Feature #5:** Record IP / confidentiality / NDA requirements more thoroughly
- **Feature #6:** New Role for University Relationship Manager (Future Cities Specific)
- **Feature #7:** Term offering import (for active term's courses, instructors, held-with)
 - Similar to how UWaterloo does it with Crowdmark (more info [here](#))





SE 490 - F26 Pilot Plan

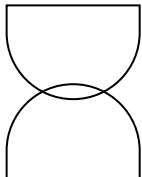


Pilot Scope

- Configure WatMatch (single-course) for SE 490 Fall 2026
- Give Jamieson + 4 WatMatch members full **admin** access to run the platform
- Give Kami **instructor** access for SE 490 + onboard SE 490 **students** onto the platform:
 - Import their capstones from SE 390
 - Create new capstone ideas for SE 490
- Give Derek Rayside **University Mentor & External Partner** access to pilot the workspace
- Give Angie Hildebrand **Academic Advisor (Community Mentor)** access for the workspace

Notes

- Until multi-role switching exists, anyone holding two roles should receive two accounts
- Pilot rollout plan is subject to change depending on Prof. Rayside's VM request + ADFS





Thank You

That's a wrap for Spring 2026!

